

Analyzing Protein–Protein Coupled Motions to Understand the Conformational Dynamics of Protein Complexes

Protein function depends on motions that are difficult to infer from a single structure. Binding a partner or forming an oligomer can change how domains move together, even when the overall fold remains similar. In this talk, I will show how comparative simulations identify these changes in two protein systems. First, I will discuss bHLH–PAS transcription factor complexes. Their conserved domain architecture allows us to compare motions across different binding partners. Using elastic network models and molecular dynamics, we separate motions within individual domains from those directly coupled to their partners, revealing differences among complexes. I will then examine leukotriene A4 hydrolase, where all-atom simulations show that dimerization does not necessarily increase conformational stability. Instead, interchain interactions alter coupling between the active site and C-terminal domain and attenuate the dynamic restraint induced by the inhibitor bestatin. Across both studies, protein–protein coupled-motion analysis helps identify how interactions between protein chains reshape collective motions.